

Lab sheet-8

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```
import os
```

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
```

```
from sklearn.metrics import (
```

```
    accuracy_score,
```

```
    precision_score,
```

```
    recall_score,
```

```
    f1_score,
```

```
    confusion_matrix,
```

```
    classification_report
```

```
)
```

```
# Step 0: Setup Output Directory
```

```
output_dir = "churn_output_images"
```

```
os.makedirs(output_dir, exist_ok=True)
```

```
# Step 1 & 2: Load Dataset from GitHub
```

```
DATA_URL = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/master/data/Telco-Customer-Churn.csv"
```

```
try:
```

```
    df = pd.read_csv(DATA_URL)
```

```
    print("Dataset loaded successfully.")
```

```
except Exception as err:
```

```
    print(f"Failed to fetch dataset: {err}")
```

```
    raise SystemExit
```

```
print(f"Dataset Dimensions: {df.shape}")
```

```

print(df.head(3))

# Step 3: Data Preprocessing & Feature Encoding

# Drop customer identifier
df = df.drop(columns=['customerID'])

# Convert TotalCharges from string to numeric (handles blank spaces as NaN)
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'].str.strip(), errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(df['TotalCharges'].median())

# Map binary target variable
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})

# Identify binary columns vs. multi-class categorical columns
binary_cols = ['Partner', 'Dependents', 'PhoneService', 'PaperlessBilling']
for col in binary_cols:
    df[col] = df[col].map({'Yes': 1, 'No': 0})

# Map gender
df['gender'] = df['gender'].map({'Male': 1, 'Female': 0})

# One-Hot Encode remaining categorical variables
categorical_cols = [
    'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup',
    'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies',
    'Contract', 'PaymentMethod'
]

df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

# Separate features (X) and target (y)
X = df.drop(columns=['Churn'])
y = df['Churn']

# Step 4: Train-Test Split (80:20 Ratio)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y

```

```

)

print(f"Training Samples: {X_train.shape[0]} | Testing Samples: {X_test.shape[0]}")

# Step 5: Train Decision Tree Classifier

# Using max_depth=4 to balance expressiveness with interpretability and avoid overfitting
dt_classifier = DecisionTreeClassifier(
    criterion='gini',
    max_depth=4,
    min_samples_split=20,
    min_samples_leaf=10,
    random_state=42
)

dt_classifier.fit(X_train, y_train)

# Step 6 & 7: Model Predictions & Performance Evaluation

y_pred = dt_classifier.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("\n" + "=" * 55)

print("MODEL PERFORMANCE METRICS")

print("=" * 55)

print(f"Accuracy : {accuracy:.4f}")
print(f"Precision : {precision:.4f}")
print(f"Recall   : {recall:.4f}")
print(f"F1-Score  : {f1:.4f}")

print("\nDetailed Classification Report:")

print(classification_report(y_test, y_pred, target_names=['Retained (0)', 'Churned (1)']))

```

```

# Plot and Save Confusion Matrix

plt.figure(figsize=(6, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Retained', 'Churned'],
    yticklabels=['Retained', 'Churned']
)

plt.title('Confusion Matrix - Customer Churn', fontsize=12, fontweight='bold')
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.tight_layout()

cm_path = os.path.join(output_dir, "confusion_matrix.png")
plt.savefig(cm_path, dpi=300, bbox_inches='tight')
print(f"[Saved Image] Confusion matrix saved to: {cm_path}")
plt.show()

# Step 8: Visualize the Decision Tree Structure

plt.figure(figsize=(24, 12))

plot_tree(
    dt_classifier,
    feature_names=X.columns.tolist(),
    class_names=['Retained', 'Churned'],
    filled=True,
    rounded=True,
    fontsize=9
)

```

```

plt.title('Trained Decision Tree (max_depth=4)', fontsize=15, fontweight='bold')
plt.tight_layout()
tree_path = os.path.join(output_dir, "decision_tree_structure.png")
plt.savefig(tree_path, dpi=300, bbox_inches='tight')
print(f"[Saved Image] Decision tree visualization saved to: {tree_path}")
plt.show()

# Step 9: Feature Importance Analysis
importance_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': dt_classifier.feature_importances_
}).sort_values(by='Importance', ascending=False)

# Keep only top 8 features contributing to splits
top_features = importance_df[importance_df['Importance'] > 0.01]

plt.figure(figsize=(8, 4.5))
sns.barplot(
    data=top_features,
    x='Importance',
    y='Feature',
    palette='viridis'
)

plt.title('Key Features Driving Customer Churn', fontsize=12, fontweight='bold')
plt.xlabel('Feature Importance Score')
plt.ylabel('Attribute')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
fi_path = os.path.join(output_dir, "feature_importance.png")
plt.savefig(fi_path, dpi=300, bbox_inches='tight')
print(f"[Saved Image] Feature importances saved to: {fi_path}")

```

```
plt.show()

# Step 10: Print Strategic Business Recommendations

print("\n" + "=" * 55)

print("BUSINESS RETENTION RECOMMENDATIONS")

print("=" * 55)

print("1. Target Month-to-Month Contracts: Contract type is the strongest split. Transition customers toward annual contracts with incentives.")

print("2. Monitor Tenure Milestones: Churn risk drops sharply as customer tenure passes 12 months; provide dedicated onboarding support in months 1-6.")

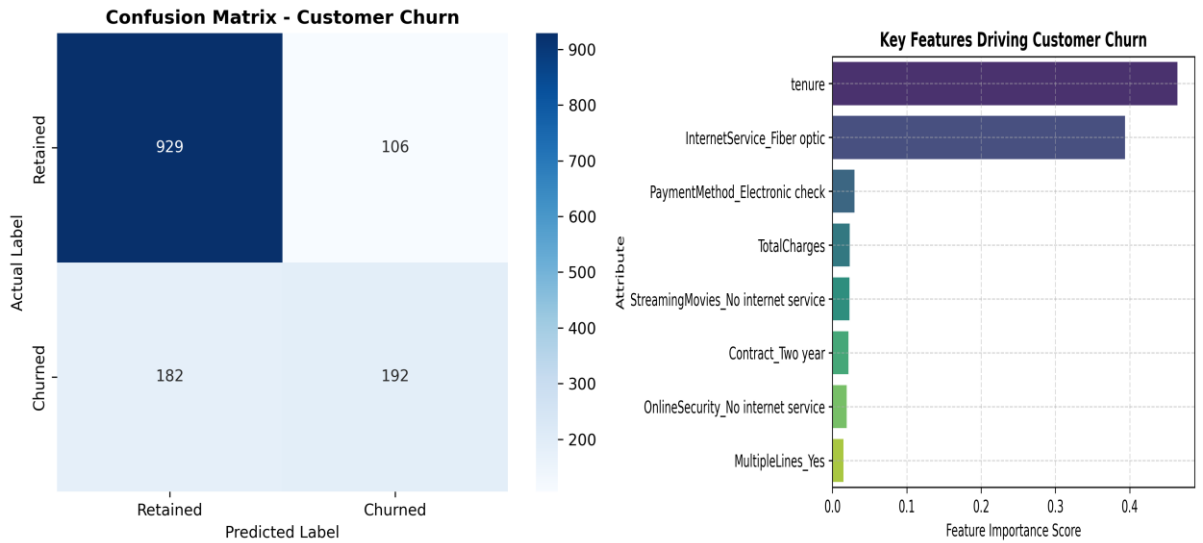
print("3. TechSupport & Fiber Quality: Customers on Fiber Optic without TechSupport exhibit high churn; bundle assistance and resolve network latency.")
```

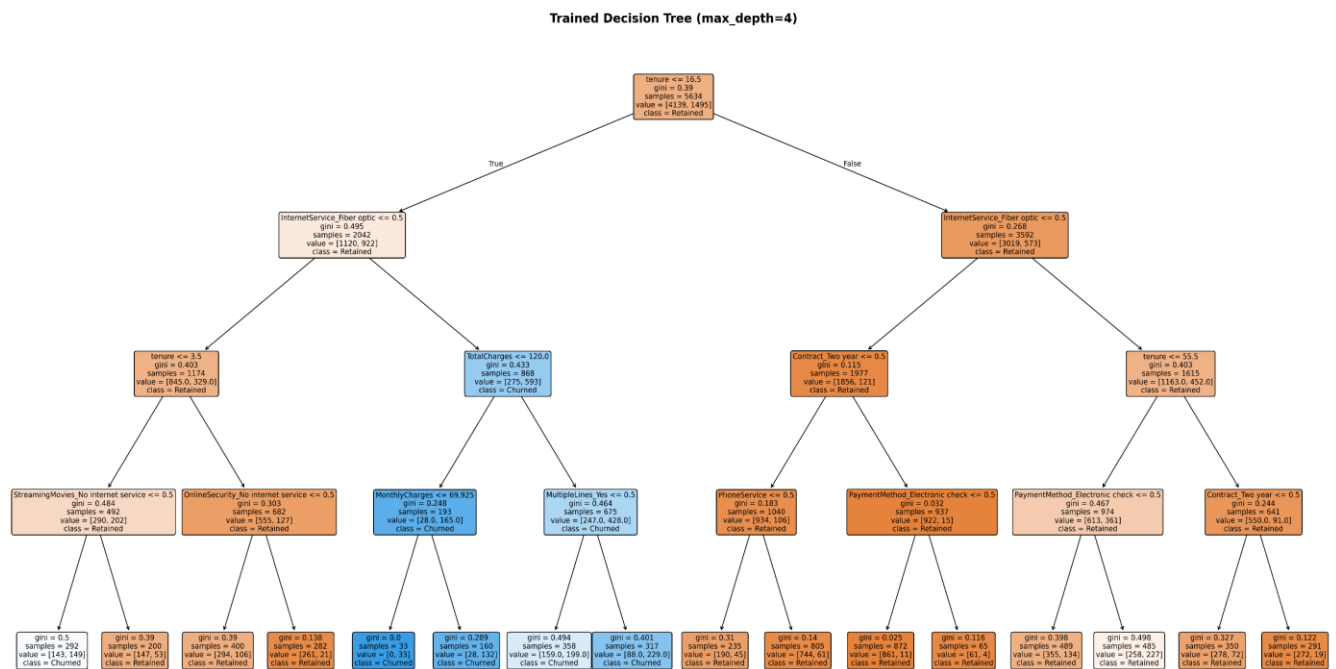
```
Dataset loaded successfully.
Dataset Dimensions: (7043, 21)
| customerID  gender  SeniorCitizen  ...  MonthlyCharges  TotalCharges  Churn
0  7590-VHVEG  Female                0  ...           29.85           29.85   No
1  5575-GNVDE  Male                  0  ...           56.95          1889.5   No
2  3668-QPYBK  Male                  0  ...           53.85          108.15  Yes

[3 rows x 21 columns]
Training Samples: 5634 | Testing Samples: 1409

=====
MODEL PERFORMANCE METRICS
=====
Accuracy   : 0.7956
Precision  : 0.6443
Recall     : 0.5134
F1-Score   : 0.5714

Detailed Classification Report:
|      |      | precision | recall | f1-score | support |
|-----|-----|-----|-----|-----|-----|
| Retained (0) |      | 0.84      | 0.90   | 0.87     | 1035    |
| Churned (1)  |      | 0.64      | 0.51   | 0.57     | 374     |
|-----|-----|-----|-----|-----|-----|
| accuracy      |      |           |         | 0.80     | 1409    |
| macro avg     |      | 0.74      | 0.71   | 0.72     | 1409    |
| weighted avg  |      | 0.79      | 0.80   | 0.79     | 1409    |
```





Experiment Questions

1. What is classification, and how does it differ from regression?

- **Classification:** A supervised learning task where the target output variable y is discrete and categorical (e.g., predicting whether a customer will churn or stay: $y \in \{0,1\}$).
- **Regression:** A supervised learning task where the target output variable y is continuous and numeric (e.g., predicting customer lifetime value, tenure duration in days, or revenue).

2. Explain the working principle of the Decision Tree algorithm.

The Decision Tree builds a decision model top-down through recursive binary splitting:

1. It scans all candidate features and evaluates every possible numerical or categorical split threshold.
2. It chooses the split that yields the largest reduction in node impurity (maximizing Information Gain or Gini reduction).
3. The dataset is partitioned into subsets at internal decision nodes.
4. This splitting repeats recursively until a stopping condition is reached (e.g., reaching max_depth, node purity, or a minimum sample threshold).
5. Leaf nodes assign predictions based on the majority class among the training points landing in that terminal region.

3. Differentiate between Gini Index and Entropy as splitting criteria.

Attribute	Gini Index (Impurity) PDF	Entropy PDF
Formula	$1 - \sum_{i=1}^c p_i^2$	$-\sum_{i=1}^c p_i \log_2(p_i)$
Computation Speed	Faster, as it involves simple arithmetic squaring.	Slower, as it requires computing logarithms for every candidate split.
Output Range	Bounds between 0.0 (pure) and 0.5 (equal binary split).	Bounds between 0.0 (pure) and 1.0 (equal binary split).
Sensitivity	Slightly more sensitive to isolating the largest class into its own node.	Tends to generate slightly more balanced tree structures.

4. What is a Confusion Matrix? Explain its components.

A Confusion Matrix is a tabular layout that compares a classification model's predicted labels against ground-truth labels across four quadrants:

- **True Positive (TP):** The model correctly predicted a churning customer as Churned.
- **True Negative (TN):** The model correctly predicted a retained customer as Retained.
- **False Positive (FP / Type I Error):** The model predicted Churned, but the customer actually stayed.
- **False Negative (FN / Type II Error):** The model predicted Retained, but the customer actually churned.

5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$ — Ratio of total correct predictions to all samples.
- **Precision:** $\frac{TP}{TP+FP}$ — The proportion of predicted churners that actually churned.
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ — The proportion of actual churners correctly detected by the model.
- **F1-Score:** $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$ — Harmonic mean of Precision and Recall.

- **Importance:** In churn detection, datasets are class-imbalanced. Relying only on accuracy allows a naive model that predicts "No Churn" for everyone to score high accuracy while missing every churner; tracking Recall and F1-Score ensures actual churn events are detected.

6. What is overfitting in a Decision Tree? How can it be reduced?

- **Overfitting:** Occurs when a decision tree grows without constraints, splitting until it memorizes noise and sample-specific idiosyncrasies of the training data, leading to poor generalization on unseen test data.
- **How to Reduce It:**
 - **Pre-pruning (Hyperparameter Constraints):** Set limits on max_depth, require a minimum number of samples to split (min_samples_split), or enforce minimum samples per leaf (min_samples_leaf).
 - **Post-pruning (Cost-Complexity Pruning):** Grow a full tree and prune sub-trees back using a cost-complexity parameter (α).
 - **Ensemble Methods:** Use bagging or boosting ensembles like Random Forests or Gradient Boosted Trees.

7. Why is customer churn prediction important for businesses?

Acquiring new customers involves high marketing and sales acquisition costs (CAC) compared to maintaining an existing account. Early churn prediction provides businesses with an operational window to proactively resolve customer dissatisfaction, offer customized discounts, and preserve recurring revenue streams before the contract is officially terminated.

8. What are the advantages and limitations of the Decision Tree algorithm?

- **Advantages:**
 - High interpretability: Rules can be visualized and explained directly to business stakeholders.
 - Requires minimal data preparation: Inherently handles both numerical and categorical variables without mandatory feature normalization.
 - Captures non-linear interactions between attributes naturally.
- **Limitations:**
 - High variance: Small changes in training data can generate an entirely different tree structure.
 - Susceptible to overfitting when unpruned.

- Orthogonal decision boundaries: Trees partition feature space strictly parallel to coordinate axes, struggling with diagonal decision boundaries.

9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

- **Credit Risk Assessment & Loan Default Approval:** Classifying whether loan applicants pose low or high credit default risk based on credit scores, income, and debt ratios.
- **Medical Diagnosis:** Assisting doctors in classifying whether a tumor is benign or malignant, or detecting diabetes risk from metabolic indicators.
- **Fraud Detection:** Identifying fraudulent credit card transactions based on transaction frequency, amount thresholds, and geographic discrepancies.

10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

- **Targeted Interventions:** Identifies high-risk customer cohorts (e.g., month-to-month contracts with fiber optic service), allowing retention teams to deploy proactive concessions instead of blanket discounts.
- **Resource Optimization:** Focuses customer service spend and retention incentives only on at-risk accounts with high lifetime value rather than wasting budget on satisfied customers.
- **Product & Service Remediation:** Highlights underlying operational failures (e.g., if lacking tech support consistently drives churn, leadership can improve technical support staffing and onboarding guides).